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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Cyber Security) (CBS) Degree Examinations

May - June 2024

22CBS201 – FIREWALL & UTM ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

	Unit – I	Marks	BT*	CO*	PO*
1. a)	What are ten commonly used network security tools and software? How do they function to enhance network security?	10	L*2	1	1
b)	Outline the functionalities of Firewalls, focusing on any five key aspects.	10	2	1	2
2. a)	Explain the different types of Firewalls and classify them based on their design and intended use.	10	2	1	2
b)	What are the reasons for implementing firewalls, and what are the potential risks associated with not having them? Please provide an explanation for each risk.	10	2	1	2
	Unit – II				
3. a)	Discuss the concept of Stateful Firewalls by highlighting their advantages and disadvantages.	10	2	2	1
b)	Distinguish between stateless and stateful firewalls, highlighting their respective functionalities and features.	10	2	2	2
4. a)	What is Unified Threat Management (UTM), and what are the security technologies and services incorporated within UTM systems?	10	2	2	2
b)	Describe the functionalities of the following: i. Content Filtering ii. Web Application Firewall	10	2	2	2
	Unit – III				
5. a)	Define Intrusion Detection Systems (IDS) and provide a brief overview of the various types of IPS.	10	2	3	2
b)	What are the different types of security architectures being utilized today? Why is the Unified Threat Management (UTM) architecture considered superior to other existing architectures? Justify	10	2	3	2
6. a)	Define URL Filtering and enumerate the various types of URL filtering methods.	10	2	3	2
b)	How do Network Intrusion Prevention Systems (NIPS), Host Intrusion Prevention Systems (HIPS), and Wireless Intrusion Prevention Systems (WIPS) differentiate from each other within Intrusion Prevention Systems (IPS)?	10	2	3	2
	Unit – IV				
7. a)	What are the features of Next-Generation Firewall (NGFW)? Discuss five advantages of NGFW.	10	2	4	2
b)	Explore different best practices involved in firewall deployment.	10	2	4	2
8. a)	Outline and elaborate on the steps for conducting generic troubleshooting.	10	2	4	2
b)	Enumerate and elaborate on five features of Next-Generation Firewalls (NGFW).	10	2	4	2
	Unit – V				
9. a)	What constitutes Threat Intelligence? Enumerate the advantages and characteristics of threat intelligence in the realm of security.	10	2	5	2
b)	What is the definition of Advanced Threat Protection (ATP)? Explain the mechanisms involved in ATP detection.	10	2	5	2
10. a)	Explain the security mechanisms of Threat Extraction and Threat Emulation.	10	2	5	2
b)	Explain the role and significance of Anti-Bot and Threat Protection mechanisms within cybersecurity frameworks.	10	2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
